1 Model

We develop a static general equilibrium model of multinational production in which the strategic agent is the ownership group. A group can operate plants in several countries, chooses which of its plants serve each market, and, when it sets quantities, takes into account that its own plants compete with one another. Markups therefore depend on the group's total share of the market rather than on the share of any single plant. Profits are paid to the countries that own each group, and ownership is recorded group by group rather than country by country. Proofs are in the Appendix.

1.1 Model Set-Up

Agents. There is a finite set of countries $\mathcal{N} = \{1, \dots, N\}$ and a finite set of sectors $\mathcal{K} = \{1, \dots, K\}$. Each country n is inhabited by a representative household that supplies L_n units of labor inelastically and owns shares in firms. Production takes place in plants, and every plant belongs to exactly one group $g \in \mathcal{G}$, which we identify with its global ultimate parent. Each group has a headquarters country $h(g)$ and operates in a single sector $k(g)$.[1] Groups are of two types. A *parent group* $g \in \mathcal{G}^P$ is headquartered in a country of a set $\mathcal{N}^P \subseteq \mathcal{N}$ and has one potential plant in every country of $\mathcal{N}^P$, its headquarters country included. A *stand-alone firm* $g \in \mathcal{G}^S$ has a single potential plant, located in $h(g)$; countries outside $\mathcal{N}^P$ host only stand-alone firms. We index plants by i, write $g(i)$ and $\ell(i)$ for the group and the location of plant i and $k(i) = k(g(i))$ for its sector, and let $\mathcal{P}_g$ be the set of potential plants of group g. Each plant produces one differentiated variety and can sell it in any country. A *market* is a destination–sector pair (n, k). Whether a group is multinational is an equilibrium outcome: a group is multinational if its active plants are located in more than one country, an active plant located outside its group's headquarters country is a foreign affiliate, and the home plant of a multinational group is a domestic multinational. A stand-alone firm is never multinational.

Representative household. The household in country n has nested CES preferences,

$$U_n = \Big[\sum_{k \in \mathcal{K}} \beta_k^{1/\eta}\, C_{nk}^{(\eta-1)/\eta}\Big]^{\eta/(\eta-1)}, \qquad C_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} q_{i,n}^{(\sigma_k-1)/\sigma_k}\Big]^{\sigma_k/(\sigma_k-1)}, \tag{1}$$

where $\mathcal{A}_{nk}$ is the set of plants active in market (n, k), $\beta_k > 0$ with $\sum_k \beta_k = 1$, and $\sigma_k > \eta \geq 1$: varieties within a sector are closer substitutes than goods in different sectors. The case $\eta = 1$ is the Cobb–Douglas limit. The associated price indices are

$$P_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} p_{i,n}^{1-\sigma_k}\Big]^{1/(1-\sigma_k)}, \qquad P_n = \Big[\sum_k \beta_k P_{nk}^{1-\eta}\Big]^{1/(1-\eta)}. \tag{2}$$

The household spends its income X_n, allocates the share $\beta_k P_{nk}^{1-\eta}/P_n^{1-\eta}$ of it to sector k, and its welfare is real income X_n/P_n. Firms buy intermediate inputs as the same sectoral composites (below), so all purchases of sector-k varieties in country n, by households and by firms, are aggregated by C_{nk}. Writing $q_{i,n}$ for the total quantity of plant i's variety sold in n and E_{nk} for total spending on sector k in n, demand is

$$q_{i,n} = p_{i,n}^{-\sigma_k}\, P_{nk}^{\sigma_k - 1} E_{nk}. \tag{3}$$

[1] This restriction is without loss for behavior. Marginal costs are constant and fixed costs are specific to each plant–market pair, so a group's decisions separate across markets (Section 1.2). A group active in several sectors is equivalent to several single-sector groups with the same owners; what the restriction rules out is a capability shared across sectors.

Market sizes E_{nk} are determined in general equilibrium (Section 1.3).[2]

Production technology. Labor is the only primary factor. A plant in sector k located in ℓ combines labor with a bundle of intermediate inputs. Writing $h_i \equiv h(g(i))$ for the headquarters country of plant i, its unit cost of production is

$$a_i = \frac{1}{\varphi_i}\, w_\ell^{1-\nu_k}\, (P_{\ell k}^{\mathrm{IO}})^{\nu_k}\, \gamma_{h_i \ell}, \qquad P_{\ell k}^{\mathrm{IO}} = \prod_{k' \in \mathcal{K}} P_{\ell k'}^{\omega_{k'k}}, \tag{4}$$

where w_ℓ is the wage, $\nu_k \in [0,1)$ is the share of intermediates in costs, and $\omega_{k'k} \geq 0$, with $\sum_{k'} \omega_{k'k} = 1$, is the share of sector k's intermediate spending that falls on sector k' (Caliendo and Parro, 2015); the constant of the Cobb–Douglas cost function is absorbed into φ_i. The term $\gamma_{h\ell} \geq 1$, with $\gamma_{hh} = 1$, is the efficiency loss from producing away from headquarters (Ramondo and Rodríguez-Clare, 2013; Head and Mayer, 2019). Intermediates are bought in the markets (ℓ, k') of the country where the plant operates, which include imported varieties and are subject to that country's tariffs, so the entry of a plant into a market lowers the input costs of every producer located in that country.

Head-office capability. The productivity of plant i combines a location, a plant and a group component,

$$\varphi_i = z_\ell\, e^{u_i} \times \begin{cases} \exp\big[(1+\delta^P \alpha_k)\, x_g\big] & \text{if } g \in \mathcal{G}^P, \\ \exp\big[-\delta^S \alpha_k\big] & \text{if } g \in \mathcal{G}^S,\ h(g) \in \mathcal{N}^P, \\ 1 & \text{if } g \in \mathcal{G}^S,\ h(g) \notin \mathcal{N}^P, \end{cases} \tag{5}$$

where $z_\ell > 0$ is the productivity of location ℓ, common to all plants there whatever their owner; u_i is an idiosyncratic plant draw, normal with mean zero and independent across plants; $x_g \geq 0$ is the capability of parent group g, with e^{x_g} Pareto with unit lower bound and shape ϑ; and $\alpha_k \in [0,1]$ indexes the head-office intensity, or complexity, of sector k (Antràs, 2003; Antràs and Helpman, 2004). Head-office services (management, technology, brands) are a non-rival capability of the group that all of its plants share, so they enter productivity rather than the cost of inputs. A stand-alone firm has no access to them, and its productivity falls short by a factor that grows with complexity at rate $\delta^S \geq 0$. A parent group's capability is worth more in complex sectors: its group component is Pareto with shape $\vartheta/(1+\delta^P \alpha_k)$, with a thicker tail where complexity is higher. Stand-alone firms in countries outside $\mathcal{N}^P$ stand in for all supply from those countries, multinational or not, and carry no penalty. The complexity index α_k appears only in (5).[3]

Pool of potential groups. The set of potential groups is drawn once, before any decision is taken, and the equilibrium is computed for a given draw. Each country–sector pair (n,k) has a given number M_{nk}^S of potential stand-alone firms, and each sector k has a given number M_k^P of potential parent groups. The headquarters country of each potential parent group is drawn independently from $\mathcal{N}^P$ with probability proportional to $L_h z_h^\zeta$, with $\zeta \geq 0$, and its capability x_g is drawn from the same distribution whatever its headquarters. Larger and more productive countries therefore host more potential parents, and their best parents tend to be better only because more draws are taken there (Gaubert and Itskhoki, 2021).

[2] Intermediates are combined across sectors with Cobb–Douglas weights, so when they are used ($\nu_k > 0$ below) we set $\eta = 1$: final and intermediate demand then share the same upper-tier elasticity, and the markup in (10) is the one firms face. The case $\eta > 1$ is used only without intermediates.

[3] The common alternative treats head-office services as labor bought at the parent's wage, so that the labor term in (4) becomes $(w_h^{\alpha_k} w_\ell^{1-\alpha_k})^{1-\nu_k}$ (Antràs, 2003). This can overturn the substitutability of labor across countries on which uniqueness of wages rests (Appendix C).

Trade. Shipping goods from ℓ to n entails an iceberg cost $d_{\ell n} \geq 1$, with $d_{nn} = 1$. Trade costs run from the country where the good is produced, not from headquarters, so a single plant can serve several destinations, including third markets, as an export platform (Tintelnot, 2017). Country n levies a tariff $t_{\ell nk} > -1$ on sector-k goods produced in ℓ, with $t_{nnk} = 0$; negative values are import subsidies. Tariffs depend on where a good is produced, not on who owns the plant that produces it, so a country's tariffs also fall on the foreign affiliates of the groups its residents own. We model the tariff as a proportional wedge on the cost of delivering the good. The delivered marginal cost of plant i in market (n, k) is

$$c_{i,n} = a_i \, d_{\ell n} \, (1 + t_{\ell nk}), \qquad \ell = \ell(i), \tag{6}$$

and country n collects $t_{\ell nk} \, a_i d_{\ell n} q_{i,n}$ on the sales of plant i.[4]

Multinational corporations and entry. A plant that serves market (n, k) pays a fixed cost

$$F_{i,n} = f_k \, w_\ell \, f^d_{\ell n}, \tag{7}$$

in units of labor of the country where the plant is located, with $f_k > 0$ and $f^d_{\ell n} \geq 1$ raising the cost of serving distant markets. The cost is incurred once per market served, whatever the quantity sold, and stands for registration, distribution and marketing; it does not depend on who owns the plant. Because it is paid in labor, entry uses real resources, appears as wage income, and leaves the model homogeneous of degree one in wages. We abstract from a fixed cost of setting up a plant. Marginal costs are constant and every fixed cost is specific to a plant–market pair, so a group's problem separates across destinations: in each market $(n, k(g))$ the group chooses the subset $T_{g,n} \subseteq \mathcal{P}_g$ of its plants that serve n. A parent group can serve a market from a plant at home, from a plant in the destination, from a plant in a third country, or from several at once; a stand-alone firm decides only whether to serve n. A plant operates if it serves at least one market, so which affiliates exist, and where each of them sells, is an equilibrium outcome. This is a simplified version of the multi-plant location problem of Yang (2023).[5]

Profit repatriation. Country n owns a share $\theta_{gn} \geq 0$ of group g, with $\sum_n \theta_{gn} = 1$. Groups maximize their own profits whoever owns them: ownership determines where profits are spent, not how groups price or which markets they enter. The profit of a group net of fixed costs, $\Pi_g = \sum_n [\pi_{g,n} - \sum_{i \in T_{g,n}} F_{i,n}]$, where $\pi_{g,n}$ is its variable profit in market $(n, k(g))$ (Section 1.2), is paid to its owners in proportion to θ_{gn}, wherever it is earned. Tariff revenue is rebated lump sum to the household of the country that collects it. Income in country n is

$$X_n = w_n L_n + \sum_{g \in \mathcal{G}} \theta_{gn} \, \Pi_g + T_n, \qquad T_n = \sum_k \sum_{i \in \mathcal{A}_{nk}} t_{\ell(i)nk} \, a_i \, d_{\ell(i)n} \, q_{i,n}. \tag{8}$$

Profits are earned where groups sell and received where their owners reside, and θ links the two. A natural benchmark assigns each group to the country of its ultimate parent, $\theta_{gn} = \mathbf{1}\{n = h(g)\}$; nothing in the definition of equilibrium requires this form. Since a group can always choose to serve no market, its net profit is non-negative in equilibrium. Because income includes the profits of the groups a country owns, wherever they produce and sell, trade need not balance country by country.

[4] A duty on the exporter's price yields the same prices when all the plants a group uses to serve n face the same tariff. When they do not, it makes the group's markup differ across its plants, and the property that a group's markup depends only on its total share (Section 1.2) is lost.

[5] A plant-level setup cost would link a group's choices across markets and turn its problem into a combinatorial one over locations and markets jointly (Tintelnot, 2017).

Timing. The model is static. The pool of potential groups and all productivity draws are realized first. Then, simultaneously, groups choose which of their plants serve each market and how much each sells, and households choose their consumption. Groups take wages, input prices and market sizes as given. They take into account their effect on a market's price index when they set quantities, but take the competitiveness of the market as given when they choose which plants operate (Section 1.2).

1.2 Market Structure

Fix a market (n, k), take the set of plants that serve it as given until the last paragraph of this section, and suppress the market's indices: we write σ, E and P for σ_k, E_{nk} and P_{nk}, c_i for the delivered cost $c_{i,n}$, and T_g for the plants of group g that serve the market.

Demand. Let $A \equiv \sum_{i \in \mathcal{A}} p_i^{1-\sigma} = P^{1-\sigma}$. By (3), the revenue share of plant i and the share of group g are

$$s_i = \frac{p_i q_i}{E} = \frac{p_i^{1-\sigma}}{A}, \qquad S_g = \sum_{i \in T_g} s_i, \qquad \sum_g S_g = 1, \tag{9}$$

where $S_g = 0$ for a group that does not serve the market. The aggregate A summarizes how competitive the market is: it is large when rivals charge low prices, and then any given price wins a small share.

Competition. Groups are the strategic agents. Each group chooses the quantities of all of its plants that serve the market, taking as given the quantities of other groups and the overall demand for the sector, which is the level of spending E when $\eta = 1$. It maximizes its variable profit $\pi_g = \sum_{i \in T_g} (p_i - c_i) q_i$.

Proposition 1 (Markups)**.** *Under quantity competition, the first-order conditions of group g are $p_i = \mu(S_g)\, c_i$ for every $i \in T_g$, with*

$$\frac{1}{\mu(S)} = 1 - \frac{1 - S}{\sigma} - \frac{S}{\eta}. \tag{10}$$

Under price competition the markup is $\mu(S) = \varepsilon(S)/[\varepsilon(S) - 1]$ with $\varepsilon(S) = \sigma - (\sigma - \eta)S$.

Both are the markups of Atkeson and Burstein (2008), evaluated at the share of the group rather than of the plant, and both satisfy the following property whenever $\sigma > \eta$.

Assumption 1. μ is continuously differentiable and strictly increasing on $[0, 1)$, with $\mu(0) = \sigma/(\sigma - 1)$.

The results for a single market and for input prices use only Assumption 1; the others use the specific form of the markup, and we say so where they do. We take quantity competition as the baseline.[6]

[6] When $\eta = 1$ the markup (10) is unbounded as $S \to 1$, so a market served by a single group has no equilibrium. Such a configuration is never an entry equilibrium if a second group can serve the market and its spending exceeds that group's fixed cost, so we consider only configurations with at least two active groups in every market.

Cannibalization across multinational affiliates. When a group expands one of its plants, the sector price index falls and every variety in the market loses sales, the group's other plants included. A single-plant firm ignores the losses it imposes on others; a group counts the losses it imposes on itself, and these grow with its share. A group with a negligible share competes only with rivals, at the within-sector elasticity σ. A group holding the whole market is the sector, and competes only with other sectors, at the elasticity η. Equation (10) weights the two by the group's share, so the markup rises from $\sigma/(\sigma-1)$ for a negligible group toward the monopoly markup. Two implications follow. First, all plants of a group charge the same markup in a given market, and differences in their costs show up in quantities rather than in markups. Second, market power is a property of the group: concentration should be measured at the level of groups, and a group charges more than its plants would charge as independent firms. Within a group, (9) and the common markup give

$$\frac{s_i}{S_g} = \frac{c_i^{1-\sigma}}{K_g}, \qquad K_g \equiv \sum_{i \in T_g} c_i^{1-\sigma}. \tag{11}$$

We call K_g the *strength* of the group in the market: an index of its cost efficiency that rises with the number of plants serving the market and with how cheaply each delivers. Substituting $p_i = \mu(S_g)c_i$ into (9),

$$\psi(S_g) = \frac{K_g}{A}, \qquad \psi(S) \equiv S\,\mu(S)^{\sigma-1}. \tag{12}$$

Under Assumption 1 the map ψ is strictly increasing, so $S_g = \psi^{-1}(K_g/A)$: the share of a group depends only on its strength relative to the competitiveness of the market. However many plants a group runs, and in however many countries, its position in the market is summarized by one number, K_g, and all it needs to know about its rivals is A. Given the active plants, the equilibrium of the market is the value of A at which shares add up to one,

$$\sum_g \psi^{-1}\left(\frac{K_g}{A}\right) = 1. \tag{13}$$

Profits. Variable profit is $\pi_g = E\,S_g\,m(S_g)$, where $m(S) \equiv 1 - 1/\mu(S)$ is the Lerner index. Under quantity competition (10) makes the Lerner index linear in the share, $m(S) = (1+\kappa S)/\sigma$ with $\kappa \equiv \sigma/\eta - 1$, so that

$$\pi_g = \frac{E}{\sigma}(S_g + \kappa S_g^2). \tag{14}$$

Under price competition the Lerner index $1/\varepsilon(S)$ is not linear, and $\kappa = (\sigma-\eta)/\sigma$ is the slope of its first-order expansion. Profits rise more than proportionally with the share, because a larger group both sells more and earns more on each unit sold. The curvature κ governs the ownership results of Section 1.4. It is zero with constant markups. Under quantity competition the Lerner index is linear, which makes those results exact; this is the main reason for our baseline.

Entry. A group's plant set matters for its profit only through its strength. Let $K(T) = \sum_{i\in T} c_i^{1-\sigma}$ and $\Phi(T) = \sum_{i\in T} F_{i,n}$ be the strength and the fixed cost of a set $T \subseteq \mathcal{P}_g$. Given A, serving the market with T yields

$$U_g(T;A) = \Omega(K(T);A) - \Phi(T), \qquad \Omega(K;A) \equiv E\,\psi^{-1}(K/A)\,m(\psi^{-1}(K/A)). \tag{15}$$

A group with $|\mathcal{P}_g|$ potential plants chooses among $2^{|\mathcal{P}_g|}$ sets. Because Ω is strictly increasing in K, a set can be optimal only if no other set delivers at least as much strength at no greater fixed cost.

These efficient sets form a chain, ordered by strength and by cost together, so the group's choice amounts to how far to climb a ladder that runs from small and cheap to large and expensive. A stand-alone firm has a ladder with a single rung. When a group's plants pay the same fixed cost to serve the market, the efficient sets are nested: each rung adds the plant with the next-lowest delivered cost, and the group's choice is a cutoff on its own list of plants, the within-group analogue of the cutoff in Gaubert and Itskhoki (2021). When fixed costs differ across its plants, the rungs need not be nested, but the ladder remains ordered by strength and cost.

Definition 1 (Entry equilibrium in a market)**.** Given delivered costs $\{c_i\}$, fixed costs $\{F_{i,n}\}$ and spending E, an entry equilibrium is a collection of active sets $\{T_g\}$ and an aggregate A such that (i) each T_g maximizes $U_g(\cdot\,; A)$ over subsets of $\mathcal{P}_g$, and (ii) A solves (13) with $K_g = K(T_g)$.

Groups thus evaluate entry at the prevailing degree of competition, as in a free-entry condition, while taking into account their effect on A when they set quantities. The difference from a notion in which groups anticipate how their entry changes A matters only for groups with large shares. This notion turns each group's entry problem into the choice of one number given one number, and this is what delivers uniqueness in Section 1.3. Appendix B discusses the alternative.

1.3 General Equilibrium

Market sizes. Let $V_{i,n'} \equiv a_i\, d_{\ell(i)n'}\, q_{i,n'}$ be the variable cost of plant i in serving country n', exclusive of tariffs. By Proposition 1, $V_{i,n'} = r_{i,n'}/[\mu_{g(i)}(1 + t_{\ell(i)n'k})]$, where $r_{i,n'}$ is the plant's revenue in that market and $\mu_{g(i)}$ its group's markup there. Spending on market (n, k) is final spending by households plus intermediate purchases by every plant located in n, wherever it sells:

$$E_{nk} = \frac{\beta_k P_{nk}^{1-\eta}}{P_n^{1-\eta}}\, X_n + \sum_{k' \in \mathcal{K}} \omega_{kk'}\, \nu_{k'} \sum_{n' \in \mathcal{N}} \sum_{i \in \mathcal{A}_{n'k'}:\, \ell(i) = n} V_{i,n'}. \tag{16}$$

Labor market. In each country, labor demanded for production and for serving markets equals the labor supplied:

$$w_n L_n = \sum_{k \in \mathcal{K}} (1 - \nu_k) \sum_{n' \in \mathcal{N}} \sum_{i \in \mathcal{A}_{n'k}:\, \ell(i) = n} V_{i,n'} \;+ \sum_{n' \in \mathcal{N}} \sum_{i:\, \ell(i) = n,\; i \in T_{g(i),n'}} F_{i,n'}. \tag{17}$$

Definition 2 (Equilibrium)**.** An equilibrium consists of wages $\{w_n\}$, price indices $\{P_{nk}\}$, market sizes $\{E_{nk}\}$, incomes $\{X_n\}$, active sets $\{T_{g,n}\}$, and prices and quantities $\{p_{i,n}, q_{i,n}\}$ such that households maximize (1); in every market, prices satisfy Proposition 1, quantities satisfy (3), and the active sets and the aggregate form an entry equilibrium (Definition 1); costs (4)–(6) are evaluated at the price indices (2) that they generate; market sizes and incomes satisfy (16) and (8); and labor markets clear, (17).

All prices, costs, fixed costs and incomes are homogeneous of degree one in wages, and shares are homogeneous of degree zero. Since variable profits and fixed costs scale together, entry decisions are unchanged when all wages are scaled by a common factor, so wages are determined up to scale. By Walras' law one labor-market condition is implied by the others, and we set $w_1 = 1$.

Existence and uniqueness. We proceed layer by layer: one market given its active plants; entry into one market; input prices, market sizes and incomes given wages and active plants; and wages given active plants.

Proposition 2 (One market)**.** *Under Assumption 1, for any active set containing at least two groups and any positive costs, the market has exactly one equilibrium.*

The proof uses (13): each $\psi^{-1}(K_g/A)$ falls as A rises, so the sum of shares is a strictly decreasing function of a single number and equals one exactly once. Each group's profit is strictly concave in its own quantities, so the first-order conditions identify the unique best response.

Entry is harder. Each entry lowers the value of entering for everyone else, and discrete games of this kind typically have several equilibria. Ranking entrants by cost restores uniqueness only when every entrant is a separate firm, and fixing an order of moves does so only by selecting one equilibrium. We instead separate the effect of a group's strength on its own payoff from the effect of its rivals. Write Ω_K and Ω_{KA} for the partial derivatives of Ω.

Theorem 1 (Entry)**.** *Consider quantity competition with $\eta = 1$ and $\sigma \geq 2$.*

(a) $\Omega(K; A)$ is strictly concave in K at every share.

(b) $\Omega_{KA} < 0$ whenever the group's share is at most one half.

(c) Suppose that, for any two candidate equilibria, no group would hold more than half of the market if it operated the plants it chooses in one of them while facing the aggregate of the other, and that no group is indifferent between two rungs of its ladder. Then a market has at most one entry equilibrium.

Part (a) says that strength has diminishing returns for the group, and part (b) that it is worth less in a more competitive market. Together they imply that each group's optimal strength falls as A rises, while A rises with every group's strength, so two equilibria with different aggregates cannot both exist. No order of entry is imposed, and groups still internalize competition among their own plants. The hypothesis, that no group holds more than half of a market, can be checked, and it is sufficient rather than necessary: at a share of one half the expression whose sign gives (b) still equals $1/\sigma$. Numerical checks suggest that it also suffices under price competition and for $\eta > 1$, but we have no proof. Because entry is chosen in whole plants, an exact equilibrium in pure strategies need not exist (Appendix B).

Proposition 3 (Prices and incomes)**.** *Fix wages and active sets. (a) The price indices are the unique fixed point of* (2) *combined with* (4)–(6)*, and iteration converges to it at rate* $\max_k \nu_k$*. (b) If tariffs are non-negative, market sizes and incomes are the unique solution of the linear system* (8)*,* (16)*.*

Proposition 4 (Existence of wages)**.** *Fix active sets, and suppose that $\nu_k = 0$, $\eta = 1$ and tariffs are non-negative. Suppose also that (i) no set of countries is cut off from the rest: for every proper subset Z of countries, some plant located in Z is active in a market of a country outside Z; and (ii) fixed costs are small enough that every country's income is at least a fraction $\epsilon > 0$ of its labor income at all wages. Then there exist wages that clear every labor market.*

Uniqueness of wages is the last layer and the most delicate. Let $W_n(w)$ be the wage bill of country n implied by (17) at wages w, and $M_{nj} \equiv \partial \ln W_n / \partial \ln w_j$. Homogeneity makes every row of M sum to one. Labor in different countries is a gross substitute when $M_{nj} > 0$ for $n \neq j$: a rise in any other country's wage raises country n's wage bill. The next lemma is the building block, and it holds without conditions.

Lemma 1 (Cost pass-through)**.** *Fix active sets. At any wage vector, the matrix $\partial \ln P / \partial \ln w$, with rows indexed by markets, is non-negative and its rows sum to one.*

In the proof, each response is a weighted average of responses one step down: a plant's cost averages wages and input prices, and a market's price index averages the cost changes of the groups in it, with weights that variable markups change but cannot make negative.

The model's solution algorithm adjusts wages by the damped tâtonnement

$$\Psi(w)_n = w_n^{1-\lambda}\big(W_n(w)/L_n\big)^{\lambda}, \qquad \lambda \in (0,1),$$

which raises a country's wage when its labor is in excess demand and whose fixed points, up to scale, are the equilibrium wage vectors. Let $d_H(w,w') = \max_n \ln(w_n/w'_n) - \min_n \ln(w_n/w'_n)$ denote Hilbert's projective metric, which is unchanged when either vector is rescaled.

Theorem 2 (Wages)**.** *Fix active sets. Let D be a compact set of wage vectors that is convex in logarithms, and suppose that (i) $M_{nj}(w) > 0$ for all $n \neq j$ and every $w \in D$, and (ii) $\lambda[1 - M_{nn}(w)] < 1$ for every n and every $w \in D$. Then there is $\tau < 1$ such that $d_H(\Psi(w),\Psi(w')) \leq \tau\, d_H(w,w')$ for all $w, w' \in D$. Hence D contains at most one equilibrium wage vector up to scale, and if Ψ maps D into itself the algorithm converges to it from any starting point in D.*

Since every row of M sums to one, gross substitutes implies $M_{nn} < 1$. When labor demand is elastic, $M_{nn} < 0$ and the undamped map ($\lambda = 1$) is not order-preserving; condition (ii), which comes from the model, says how much damping restores the property. Uniqueness itself needs only condition (i): by a classical argument, gross substitutes on D rules out a second equilibrium in D (Appendix C.6). The next result gives conditions under which (i) holds.

Proposition 5 (Gross substitutes)**.** *Suppose $\nu_k = 0$ and $\eta = 1$, and that no group's share exceeds one half in any market under quantity competition. Then for $n \neq j$,*

$$M_{nj} \;\geq\; \bar{E}_n^{(j)} \equiv \sum_{i,n'} b_{i,n'}^{n}\, \frac{\partial \ln E_{n'k(i)}}{\partial \ln w_j}, \tag{18}$$

where $b_{i,n'}^{n}$ is the share of country n's wage bill paid by plant i out of its variable costs in country n'. Labor is a gross substitute whenever $\bar{E}_n^{(j)} > 0$.

The bound says that when production uses only local labor, a rise in a foreign wage can lower country n's wage bill only by shrinking the markets in which its plants sell; every reallocation of sales between plants and between groups favors country n. The one-half condition is the same as in Theorem 1: it signs the effect of variable markups on factor demand. With intermediates, the same bound holds provided the cost of each plant paying country n rises by no more than the smaller of its group's average cost and the market's price index, a condition that is vacuous when $\nu_k = 0$.

These results have three limits. First, each layer is unique given the layers it takes as given, which does not make the joint equilibrium of wages and active sets unique: active sets change discretely with wages, and Theorem 2 holds them fixed. Second, existence is not guaranteed everywhere: an exact entry equilibrium in whole plants may fail to exist, and with intermediate inputs we have not proved that market-clearing wages exist. Third, gross substitutes is not a theorem of the model. It can fail when head-office services are a factor input (Appendix C), and with capability alone the demand term in (18) can still turn negative, because a country whose groups lose profits when a foreign wage rises spends less. That income effect is the ownership channel the model is built to study. The hypotheses of Theorems 1 and 2 and of Proposition 5 can be checked at the computed solution, and we report these checks with the quantification.

Solution algorithm. The layers above map into nested loops. Each market is solved for A from (13); with entry, each group's best response at a given A is found by scanning its ladder. Given wages and active sets, price indices come from iterating the contraction of Proposition 3 and incomes from one linear solve, and wages are updated by the tâtonnement Ψ with a step that satisfies condition (ii). An outer loop lets the groups with the largest gains change their plant sets until no group wishes to move. This loop is not guaranteed to converge and the joint equilibrium is not known to be unique, so we solve from several starting points (Appendix E).

1.4 Comparative Statics

Incidence within a market. Consider a market with given active sets and spending, and a change in delivered costs. Let $\hat{c}_g = \sum_{i \in T_g} (s_i/S_g)\, d\ln c_i$ be the average cost change of group g, $\varepsilon_\mu(S) \equiv S\mu'(S)/\mu(S)$ the elasticity of the markup, and

$$\Lambda_g \equiv \frac{1}{1 + (\sigma - 1)\, \varepsilon_\mu(S_g)}, \qquad \varpi_g \equiv \frac{S_g \Lambda_g}{\sum_h S_h \Lambda_h}. \tag{19}$$

Under quantity competition with $\eta = 1$, $\Lambda_g = (1 - S_g)/[1 + (\sigma - 2)S_g]$, which falls from one toward zero as the share rises.

Proposition 6 (Incidence)**.** *Holding active sets and spending fixed,*

$$d\ln S_g = -(\sigma - 1)\Lambda_g (\hat{c}_g - d\ln P), \qquad d\ln P = \sum_g \varpi_g\, \hat{c}_g. \tag{20}$$

If only group j's costs change, its sales-weighted average price rises by the fraction $\rho_j = 1 - (\sigma - 1)\varepsilon_\mu(S_j)\Lambda_j(1 - \varpi_j) \in (0, 1]$ of its average cost increase $\hat{c}_j$, and every rival h gains share, $d\ln S_h = (\sigma - 1)\Lambda_h \varpi_j \hat{c}_j$, and raises its price.

In the limit of small groups, which is monopolistic competition, $\Lambda_g = 1$, $\varpi_g = S_g$ and $\rho_j = 1$. Large groups instead absorb part of a cost increase in their markup, so pass-through falls with the group's share, and since $\varpi_g < S_g$ for the larger groups, the price index moves by less than their share would suggest. Rivals are not passive either: a cost increase for one group raises their shares, markups and prices. Because tariffs depend on where goods are produced, a tariff on goods from ℓ raises $\hat{c}_g$ only by the share of group g's sales that comes from its plants in ℓ: a group able to serve the market from several countries is partly shielded, while a stand-alone exporter from ℓ bears the tariff in full.

Ownership and the level of profit income. Consider a country H with ownership shares $\theta_g \equiv \theta_{gH}$, and its profit income from a market, $\pi^H = \sum_g \theta_g \pi_g$.

Proposition 7 (Ownership-weighted concentration)**.** *Under quantity competition,*

$$\frac{\pi^H}{E} = \frac{1}{\sigma}\Big[\bar{\theta} + \kappa\big(\bar{\theta} \cdot \mathrm{HHI} + \mathrm{Cov}_S(\theta, S)\big)\Big], \tag{21}$$

where $\bar{\theta} = \sum_g S_g \theta_g$ is the sales-weighted ownership share, $\mathrm{HHI} = \sum_g S_g^2$, and $\mathrm{Cov}_S(\theta, S) = \sum_g \theta_g S_g^2 - \bar{\theta} \cdot \mathrm{HHI}$ is the covariance between ownership and share computed with sales weights. A calculation that applies $\bar{\theta}$ to total profit in the market, $\bar{\theta} \sum_g \pi_g$, misses exactly $(E\kappa/\sigma)\,\mathrm{Cov}_S(\theta, S)$.

The first term is what a model with constant markups delivers: ownership times revenue over σ. The second is an ownership-weighted Herfindahl index. Its component $\bar{\theta} \cdot \text{HHI}$ can be built from a country's ownership share and a concentration index; the covariance cannot, since it requires knowing which groups a country owns and how large each one is. Because profits are convex in shares, two countries with the same share of a market's sales earn different profits if one owns the large groups and the other the small ones. With constant markups, as under monopolistic competition, $\kappa = 0$ and a country's profit income is its ownership share times total profit, whatever the distribution of ownership across groups; that benchmark lies outside the oligopoly model, where $\kappa = \sigma/\eta - 1 > 0$. The decomposition is exact under quantity competition and a first-order approximation under price competition. It concerns variable profit, before the fixed costs that a country's groups pay to serve the market. Summed over markets with spending weights, the same algebra splits the economy-wide correction into a covariance across markets between ownership shares and concentration, and the average of the covariances within markets.

Ownership and the response of profit income. Welfare depends on how profit income responds to policy, and the response is governed by a different statistic.

Proposition 8 (Response of owned profits)**.** *Under quantity competition, holding active sets and spending fixed, a change in the delivered costs of group j alone changes country H's profit income by*

$$\frac{d\pi^H}{E} = \frac{\sigma - 1}{\sigma}\, \varpi_j\, \hat{c}_j \sum_{h \neq j} S_h \Lambda_h \Big[\theta_h(1 + 2\kappa S_h) - \theta_j(1 + 2\kappa S_j)\Big]. \tag{22}$$

A cost increase for group j alone, for example a tariff on goods from a country from which only j serves the market, reallocates sales from j to its rivals in proportion to $S_h\Lambda_h$, and each unit of share is worth $\theta(1 + 2\kappa S)$ to country H, up to the factor E/σ. The country gains when the rivals it owns are worth more to it at the margin than the group whose costs rose. The weights are $S_h\Lambda_h$ rather than S_h^2, so large groups count for less in the response than in the level, and what matters is the ownership of the affected group relative to its rivals, not the covariance that governs the level. The ownership-weighted Herfindahl index measures a country's stake in a market but is not a sufficient statistic for how that stake responds to a tariff; in general equilibrium the response also runs through wages, entry and market sizes.

Entry. The proof of Theorem 1 also gives comparative statics.

Corollary 1 (Entry responses)**.** *Under quantity competition with $\eta = 1$, the strength a group chooses is non-decreasing in spending E, holding A fixed. It is non-increasing in A, holding E fixed, provided the group's share would not exceed one half with any of its rungs at the aggregates compared.*

A more competitive market leads a group to climb fewer rungs of its ladder, and a larger market to climb more. Since every rung is a set of plants, these responses determine which affiliates operate, and through (16) and (17) they feed back into market sizes and wages.

Wages and prices. By Lemma 1, input–output links and markups propagate a wage change across price indices but cannot reverse or amplify it. Who gains from a fall in j's wage depends on ownership. The fall lowers the costs of every group with plants in j, and part of the gain accrues as profit to the owners of those groups, wherever they reside. A policy that lowers a foreign wage can therefore raise the income of the country that owns the plants located there. This is the general-equilibrium channel through which ownership matters for trade policy.

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Appendix

A The Market Game

A.1 Markups. Fix a market. With η the upper-tier elasticity, write spending as $E = D\,P^{1-\eta}$, with the shifter D taken as given by firms; $\eta = 1$ gives $E = D$. Let $\rho = (\sigma-1)/\sigma$ and $Q = (\sum_i q_i^\rho)^{1/\rho}$. Inverse demand is $p_i = D^{1/\eta} Q^{1/\sigma - 1/\eta} q_i^{-1/\sigma}$, so group g's revenue is $R_g = D^{1/\eta} Q^{1/\sigma-1/\eta} \sum_{j\in T_g} q_j^\rho$. Using $\partial \ln Q/\partial q_i = q_i^{\rho-1}/Q^\rho$ and $s_i = q_i^\rho/Q^\rho$,

$$\frac{\partial R_g}{\partial q_i} = p_i\Big[\rho + \Big(\frac{1}{\sigma} - \frac{1}{\eta}\Big) S_g\Big] = p_i\Big[1 - \frac{1-S_g}{\sigma} - \frac{S_g}{\eta}\Big].$$

Setting marginal revenue equal to c_i gives (10); the bracket is common to all $i \in T_g$ and depends only on S_g. Under price competition, demand is $q_i = p_i^{-\sigma} P^{\sigma-\eta} D$, so $\partial \ln q_j/\partial \ln p_i = -\sigma \mathbf{1}\{i = j\} + (\sigma-\eta) s_i$. The first-order condition for p_i is $q_i - \sigma(p_i - c_i) q_i/p_i + (\sigma-\eta)(s_i/p_i)\sum_{j\in T_g}(p_j - c_j) q_j = 0$. Trying $p_j = \mu c_j$ for all $j \in T_g$, so that $(p_j - c_j) q_j = (1 - 1/\mu) r_j$ and $\sum_{j\in T_g} r_j = S_g E$, it reduces to $(1 - 1/\mu)[\sigma - (\sigma-\eta) S_g] = 1$, which is the stated markup.

A.2 Proof of Proposition 2. *The share map.* $\psi(S) = S\mu(S)^{\sigma-1}$ is the product of S and of $\mu(S)^{\sigma-1}$, both positive and strictly increasing on $(0,1)$ under Assumption 1, and $\psi(0) = 0$; so ψ has a continuous, strictly increasing inverse. *Best responses.* Given rivals' quantities, group g's profit is $D^{1/\eta} X_g (X_g + B_g)^e - \sum_{i\in T_g} c_i q_i$, where $X_g = \sum_{i\in T_g} q_i^\rho$, B_g is the same index for rivals, and $e = (\eta-\sigma)/[\eta(\sigma-1)] \in [-1, 0)$. The function $X \mapsto X(X+B)^e$ is strictly increasing and strictly concave for $B > 0$: its derivatives are $(X+B)^{e-1}[(1+e)X + B] > 0$ and $e(X+B)^{e-2}[(1+e)X + 2B] < 0$. The index X_g is strictly concave in $(q_i)_{i\in T_g}$. A strictly increasing, strictly concave function of a strictly concave function is strictly concave, and cost is linear, so profit is strictly concave in the group's own quantities. Since $\partial X_g/\partial q_i \to \infty$ as $q_i \to 0$, the optimum is interior and is the unique solution of the first-order conditions. Under price competition, dividing the first-order condition of A.1 by q_i gives $1 - c_i/p_i = [1 + (\sigma-\eta)\pi_g/E]/\sigma$, whose right side does not depend on i, so every interior critical point has a common markup μ. With a common markup, $S_g(\mu) = \mu^{1-\sigma} K_g/(\mu^{1-\sigma} K_g + B_g')$, where B_g' is the sum of $p^{1-\sigma}$ over rivals, and $E = DP^{1-\eta}$, so that

$$\frac{d\ln \pi_g}{d\ln \mu} = \frac{1}{\mu - 1} - \big[\varepsilon(S_g(\mu)) - 1\big].$$

The first term falls from infinity to zero as μ rises, and the second rises because S_g falls with μ; the derivative changes sign once, from positive to negative, and the best response is unique. Pricing a plant out of the market is never optimal, since adding it at the common markup raises the group's share without lowering its margin. *The aggregate.* By Proposition 1 and (12), an equilibrium is an A solving (13), and conversely any such A generates prices $p_i = \mu(S_g) c_i$ that satisfy every group's first-order conditions and are consistent with A, since $\sum_i p_i^{1-\sigma} = \sum_g \mu(S_g)^{1-\sigma} K_g = A \sum_g S_g = A$. The left side of (13), $\mathcal{S}(A)$, is continuous and strictly decreasing. As $A \to \infty$ every share tends to zero. Let $\bar\psi = \lim_{S\to 1} \psi(S)$, infinite when $\eta = 1$; as A falls toward $\max_g K_g/\bar\psi$ the largest group's share tends to one while every other share stays positive, so $\mathcal{S}$ exceeds one. Hence $\mathcal{S}(A) = 1$ has exactly one solution, and every other variable is a function of it. □

B Entry

B.1 Two elasticities. Let $G(S) \equiv \pi'(S)/\psi'(S)$, where $\pi(S) = ES\,m(S)$, and define $\varepsilon_G = SG'/G$ and $\varepsilon_\psi = S\psi'/\psi$. Differentiating $\psi(S) = K/A$ gives $\partial S/\partial K = 1/(A\psi')$ and $\partial S/\partial A = -\psi/(A\psi')$.

Hence $\Omega_K = G(S)/A > 0$, $\Omega_{KK} = G'(S)/(A^2\psi'(S))$, and

$$\Omega_{KA} = -\frac{1}{A^2}\Big[G'(S)\frac{\psi(S)}{\psi'(S)} + G(S)\Big] = -\frac{G(S)}{A^2\varepsilon_\psi(S)}\big[\varepsilon_G(S) + \varepsilon_\psi(S)\big].$$

So Ω is strictly concave in K if and only if $\varepsilon_G < 0$, and $\Omega_{KA} < 0$ if and only if $\varepsilon_G + \varepsilon_\psi > 0$.

B.2 Proof of Theorem 1(a)–(b). With $\eta = 1$, $\mu(S) = \sigma/[(\sigma-1)(1-S)]$. Set $E = 1$, $a \equiv 2(\sigma-1)$ and $b \equiv \sigma - 2 \geq 0$. Then $\pi'(S) = (1+aS)/\sigma$ and $\psi'(S) = C(1-S)^{-\sigma}(1+bS)$ with $C = [\sigma/(\sigma-1)]^{\sigma-1}$, so

$$\varepsilon_G(S) = S\,f(S), \qquad f(S) = \frac{a}{1+aS} - \frac{b}{1+bS} - \frac{\sigma}{1-S}.$$

At $S = 0$, $f(0) = a - b - \sigma = 0$. Since $a > b \geq 0$, $a/(1+aS) > b/(1+bS) \geq 0$, so $f'(S) = -a^2/(1+aS)^2 + b^2/(1+bS)^2 - \sigma/(1-S)^2 < 0$. Hence $f < 0$ and $\varepsilon_G < 0$ on $(0,1)$, which is (a). Next, $\varepsilon_\psi = (1+bS)/(1-S)$, and using $1 + bS - \sigma S = 1 - 2S$,

$$\varepsilon_G + \varepsilon_\psi = \frac{aS}{1+aS} - \frac{bS}{1+bS} + \frac{1-2S}{1-S}.$$

The map $x \mapsto x/(1+x)$ is increasing and $a > b$, so the first two terms sum to a positive number; the third is non-negative when $S \leq \frac{1}{2}$. This is (b). At $S = \frac{1}{2}$ the expression equals $(\sigma-1)/\sigma - (\sigma-2)/\sigma = 1/\sigma$. □

B.3 Proof of Theorem 1(c). *Step 1: the ladder.* Since $\Omega_K > 0$, a set T is optimal only if no T' has $K(T') \geq K(T)$ and $\Phi(T') \leq \Phi(T)$ with one inequality strict. The remaining efficient sets can be ordered so that K and Φ rise together. *Step 2: single crossing.* Take $A < A'$ and two rungs with $K_1 < K_2$, and let $\Delta(x) = \int_{K_1}^{K_2} \Omega_K(K;x)\,dK - [\Phi_2 - \Phi_1]$ be the net gain from the larger rung at aggregate x. The share $\psi^{-1}(K/x)$ rises with K and falls with x, so on $[K_1, K_2] \times [A, A']$ it is largest at (K_2, A). If $\psi^{-1}(K_2/A) \leq \frac{1}{2}$, part (b) gives $\Omega_{KA} < 0$ throughout, and Δ is strictly decreasing on $[A, A']$. *Step 3: the aggregate.* Given strengths, the aggregate $A(\{K_g\})$ solving (13) is strictly increasing in each K_g: raising K_j at a given A raises the left side, which is strictly decreasing in A. *Step 4.* Suppose $(A, \{K_g\})$ and $(A', \{K'_g\})$ are equilibria with $A < A'$, and suppose $K'_g > K_g$ for some g. Apply Step 2 with $K_1 = K_g$ and $K_2 = K'_g$: by the hypothesis, group g would hold at most half of the market with the plants it chooses at A' while facing A, so $\psi^{-1}(K'_g/A) \leq \frac{1}{2}$ and Δ is strictly decreasing on $[A, A']$. Optimality at A' gives $\Delta(A') \geq 0$, hence $\Delta(A) > 0$, so at A the group strictly prefers K'_g to K_g, contradicting optimality at A. Hence $K'_g \leq K_g$ for every g (the single-crossing argument of Milgrom and Shannon, 1994), and by Step 3, $A' = A(\{K'_g\}) \leq A(\{K_g\}) = A$, a contradiction. Equilibria with the same aggregate have the same best responses, and with no ties the same active sets. □

B.4 The notion of entry equilibrium and existence. Definition 1 evaluates every alternative set at the prevailing aggregate A. In a two-stage game in which the market is re-solved after a deviation, a group that adds a plant anticipates that A rises, which lowers the value of the deviation; the two notions coincide for groups with negligible shares and can differ for large ones. Theorem 1 applies to the notion of Definition 1. For the two-stage notion we have only numerical evidence, from markets small enough to enumerate every configuration and re-solve the market after every deviation. Existence in pure strategies is not guaranteed under either notion: a group's choice is discrete, so $K_g^*(A)$ jumps, and (13) may have no solution with each group on a rung. The algorithm of Section 1.3 then returns the configuration with the fewest groups wishing to deviate.

C General Equilibrium

C.1 Proof of Proposition 3. *(a)* Let T map log price indices $\ln P$ into the log price indices generated by the costs (4)–(6) that P implies, holding wages and active sets fixed. Multiplying every cost in a market by a constant leaves shares and markups unchanged and multiplies every price by that constant, so each market's price index is homogeneous of degree one in the costs entering it; by Lemma 1's second step it is also non-decreasing in each of them. In (4), $\ln a_i$ depends on $\ln P$ through $\nu_k \sum_{k'} \omega_{k'k} \ln P_{\ell k'}$, with non-negative weights summing to ν_k. Hence T is monotone and $T(\ln P + x\mathbf{1}) \leq T(\ln P) + (\max_k \nu_k)\, x\mathbf{1}$ for $x \geq 0$, so by Blackwell's sufficient conditions T is a contraction in the sup norm with modulus $\max_k \nu_k < 1$. *(b)* Given shares and markups, the second term of (16) is linear in market sizes, $E = \mathcal{E}X + BE$, where $B \geq 0$ and the column of B for market (n', k') sums to $\nu_{k'} \sum_i s_i/[\mu(1+t)]$. This is below $\nu_{k'} < 1$ when tariffs are non-negative, since markups exceed one. So $\rho(B) < 1$ and $E = (I - B)^{-1}\mathcal{E}X \geq 0$. Variable profits and tariff revenue are linear in E and, with non-negative tariffs, non-negative. Let Φ_g be group g's total fixed costs. Then (8) reads $X = wL - \theta'\Phi + \mathcal{M}X$ with $\mathcal{M} \geq 0$. Summing over countries, $\mathbf{1}'\mathcal{M}X$ equals world spending minus intermediate purchases minus the variable wage bill, that is world final spending minus the variable wage bill; since every unit of spending generates a positive variable wage bill when $\nu_k < 1$, each column of $\mathcal{M}$ sums to less than one, $\rho(\mathcal{M}) < 1$, and $X = (I - \mathcal{M})^{-1}(wL - \theta'\Phi)$ is the unique solution. With import subsidies the system is still linear, and it has a unique solution whenever $I - B$ and $I - \mathcal{M}$ are non-singular, which can be checked directly. □

C.2 Proof of Proposition 4. Normalize wages to the unit simplex and let $z_n(w) = W_n(w)/w_n - L_n$ be excess demand for labor. By Propositions 2–3, z is continuous on the interior of the simplex and homogeneous of degree zero. It satisfies Walras' law $\sum_n w_n z_n(w) = 0$, which follows from adding (8) across countries and using (16) and the definition of W_n as the right side of (17). It is bounded below by $-L$. It remains to check the boundary condition of Mas-Colell, Whinston, and Green (1995, Proposition 17.C.1): if $w^m \to \bar{w}$, where $Z = \{n : \bar{w}_n = 0\}$ is non-empty, then $\max_n z_n(w^m) \to \infty$.

With $\nu_k = 0$, a plant's delivered cost is proportional to the wage where it is located. So the costs of plants located in Z go to zero, while all other costs stay in a compact subset of $(0, \infty)$. By (i), some plant located in Z is active in a market (n', k) with $n' \notin Z$. By (ii) and $\eta = 1$, $E_{n'k} = \beta_k X_{n'} \geq \beta_k \epsilon w_{n'} L_{n'}$, which stays bounded away from zero. For m large, the lowest-cost plant i in that market is located in some $\ell \in Z$. Let g be its group and J the number of plants in the market. Its variable cost bill is $V_i = E_{n'k} s_i/[\mu_g(1+t)]$, all of it paid to labor in ℓ, so it suffices to show that $V_i/w_\ell \to \infty$.

If $S_g \leq \frac{1}{2}$, then $\mu_g \leq \bar{\mu} \equiv \mu(\frac{1}{2})$. Every price is at least c_i, since markups exceed one, so $P^{1-\sigma} \leq J c_i^{1-\sigma}$ and $s_i = (\mu_g c_i/P)^{1-\sigma} \geq \bar{\mu}^{1-\sigma}/J$. Then V_i is bounded away from zero while $w_\ell \to 0$.

If $S_g > \frac{1}{2}$, every rival h has $S_h < \frac{1}{2}$, so $\psi(S_h) \leq \bar{\mu}^{\sigma-1} S_h \leq \bar{\mu}^{\sigma-1}(1 - S_g)$, while $\psi(S_g) \geq \frac{1}{2}\mu_g^{\sigma-1}$. Under both conducts with $\eta = 1$, $\mu(S)(1-S) \leq \sigma/(\sigma-1)$. Together with (12), this gives $K_h/K_g = \psi(S_h)/\psi(S_g) \leq C\mu_g^{-\sigma}$ for a constant C. Rival strength K_h is bounded below, because all costs are bounded above. Plant i is the cheapest in its group, so $K_g \leq J c_i^{1-\sigma}$ and $s_i \geq S_g/J > 1/(2J)$. Hence $V_i \geq c\, c_i^{(\sigma-1)/\sigma}$ for some $c > 0$, and since c_i is proportional to w_ℓ, V_i/w_ℓ grows like $w_\ell^{-1/\sigma}$.

In either case $z_\ell \to \infty$, and the result follows from Mas-Colell, Whinston, and Green (1995, Proposition 17.C.1). □

C.3 Proof of Lemma 1. Hats denote log changes in response to a change in wages. *Costs.* Trade costs and tariffs are fixed, so by (4) and (6), $\hat{c}_i = \hat{a}_i = (1 - \nu_k)\hat{w}_\ell + \nu_k \sum_{k'} \omega_{k'k} \hat{P}_{\ell k'}$, with non-negative coefficients summing to one. *Markets.* Differentiating $S_g = \mu(S_g)^{1-\sigma} K_g/A$, with

$\hat{K}_g = (1-\sigma)\hat{c}_g$ by (11) and $\hat{A} = (1-\sigma)\hat{P}$, gives $\hat{S}_g = -(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P})$, with Λ_g, ϖ_g and $\hat{c}_g$ as defined in (19). Shares sum to one, so $\sum_g S_g \hat{S}_g = 0$, which gives $\hat{P} = \sum_g \varpi_g \hat{c}_g$ with weights $\varpi_g > 0$ summing to one. *Stacking.* Substituting costs into markets gives $\hat{P} = \Gamma\hat{w} + \mathcal{N}\hat{P}$ with $\Gamma, \mathcal{N} \geq 0$, $(\Gamma + \mathcal{N})\mathbf{1} = \mathbf{1}$, and the rows of $\mathcal{N}$ summing to $\nu_k < 1$. Hence $\rho(\mathcal{N}) < 1$, $(I - \mathcal{N})^{-1} = \sum_{s \geq 0} \mathcal{N}^s \geq 0$, and $\partial \ln P / \partial \ln w = (I - \mathcal{N})^{-1}\Gamma \geq 0$ with row sums $(I - \mathcal{N})^{-1}(\mathbf{1} - \nu) = \mathbf{1}$. □

C.4 A decomposition of factor payments. Let V_i be plant i's variable cost in a market (n', k), with tariffs fixed. Since $V_i = r_i/[\mu_g(1+t)]$, $\hat{V}_i = \hat{E} + \hat{s}_i - \hat{\mu}_g$, with $\hat{s}_i = (1-\sigma)(\hat{c}_i + \hat{\mu}_g - \hat{P})$ and $\hat{\mu}_g = \varepsilon_\mu \hat{S}_g$. Using C.3 and $(\sigma - 1) = (\sigma - 1)\Lambda_g[1 + (\sigma - 1)\varepsilon_\mu]$,

$$\hat{V}_i = \hat{E} - (\sigma-1)(\hat{c}_i - \hat{c}_g) - (1-\varepsilon_\mu)(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P}). \tag{A.1}$$

The first term is the growth of the market, the second the reallocation of sales within the group, the third the reallocation between the group and its rivals. Its coefficient is non-negative if and only if $\varepsilon_\mu \leq 1$, which under quantity competition with $\eta = 1$, where $\varepsilon_\mu = S/(1-S)$, is exactly $S_g \leq \frac{1}{2}$.

C.5 Proof of Proposition 5. Raise w_j alone, $j \neq n$. With $\nu_k = 0$ and head-office services a capability, the whole variable labor bill of a plant is paid in its location, so the plants paying country n are located in n, and their costs do not respond: $\hat{c}_i = 0$. Fixed costs (7) are paid at w_n and do not respond either. By Lemma 1, $\hat{c}_g, \hat{P} \in [0,1]$. Let $\chi = (1-\varepsilon_\mu)(\sigma-1)\Lambda_g$. The one-half condition and $\Lambda_g \leq 1$ give $0 \leq \chi \leq \sigma - 1$, so the last two terms of (A.1) equal $(\sigma-1)\hat{c}_g - \chi(\hat{c}_g - \hat{P})$. If $\hat{c}_g \geq \hat{P}$ this is at least $(\sigma-1)\hat{P}$; otherwise both parts are non-negative. In either case it is at least $(\sigma-1)\min\{\hat{c}_g, \hat{P}\} \geq 0$, so $\hat{V}_i \geq \hat{E}$. Weighting by $b^n_{i,n'}$ and adding the fixed-cost part, whose response is zero, gives (18). □

C.6 Gross substitutes and uniqueness. The following classical argument shows that condition (i) of Theorem 2 alone gives uniqueness. Suppose $M_{nj} > 0$ for $n \neq j$ at every point of a set D that is convex in logarithms. Since M is homogeneous of degree zero in wages, the same holds at every rescaling of a point of D, and the set of these rescalings is still convex in logarithms. Let $w, w^* \in D$ be non-proportional equilibria. Scale w^* so that $w^* \leq w$ with equality at some n_0, and move from w^* to w along the segment in logarithms. Every wage other than w_{n_0} weakly rises and at least one strictly rises, so W_{n_0} strictly rises while w_{n_0} does not. Then $z_{n_0}(w) > z_{n_0}(w^*) = 0$, a contradiction.

C.7 Proof of Theorem 2. In logarithms, Ψ becomes a map with Jacobian $(1-\lambda)I + \lambda M$, whose rows sum to one. Its off-diagonal entries are positive by (i) and its diagonal entries by (ii), so it is a strictly positive stochastic matrix at every point of D. In logarithms, d_H is the oscillation seminorm $\operatorname{osc}(x) = \max_n x_n - \min_n x_n$. By the mean value theorem in integral form, the difference between the images of two points is $\bar{A}$ times the difference between the points, where $\bar{A}$ is the average of the Jacobian along the segment joining them, which stays in D. The contraction coefficient of a stochastic matrix in the oscillation seminorm, $\delta(A) = \frac{1}{2}\max_{i,j} \|A_{i\cdot} - A_{j\cdot}\|_1$, is convex in A, so $\delta(\bar{A}) \leq \sup_D \delta$. For a strictly positive stochastic matrix, $\delta(A) \leq \tanh(\Delta(A)/4) < 1$, where $\Delta(A) = \max_{i,j,k,l} \ln(A_{ik}A_{jl}/A_{jk}A_{il})$ (Birkhoff, 1957; Seneta, 2006). Since D is compact and M is continuous, Δ is bounded on D, and $\tau = \sup_D \tanh(\Delta/4) < 1$ gives the contraction. A fixed point of Ψ up to scale, $\Psi(w) = cw$, has $W_n = c^{1/\lambda} w_n L_n$ for every n. Walras' law then forces $c = 1$, so it is an equilibrium. Two such points in D would contradict the contraction. If Ψ maps D into itself, the iterates converge by the contraction mapping theorem, since D is compact. □

C.8 Head-office services as a factor input. Suppose instead that the labor term in (4) is $(w_h^{\alpha_k} w_\ell^{1-\alpha_k})^{1-\nu_k}$, so that country h is paid the share α_k of the labor bill of every plant its groups run. All plants of a group report to the same headquarters, so country h receives the share α_k of the group's whole bill in the market. When the group's plants serving a market face the same tariff, as in the example below, the within-group term of (A.1) averages to zero over the group. Consider a country n whose labor income consists only of head-office payments from groups whose plants are all located in a single country j, with $\nu_k = 0$. Raising w_j raises those groups' costs by $\hat{c}_g = 1 - \alpha_k$, while the price indices of the markets they serve rise only in proportion to the groups' incidence weights. By (A.1), M_{nj} is then a weighted sum over these groups of $\hat{E} - (1-\varepsilon_\mu)(\sigma-1)\Lambda_g(\hat{c}_g - \hat{P})$. When the groups are small, $\Lambda_g \approx 1$, $\varepsilon_\mu \approx 0$ and $\hat{c}_g - \hat{P} \approx 1 - \alpha_k$, so M_{nj} is negative whenever $(\sigma-1)(1-\alpha_k)$ exceeds the growth of the markets they serve, although no group is near one half of any market. The failure is worst when α_k is small: a large head-office share ties the group's cost to its own country's wage and insulates it from w_j. What breaks gross substitutes is a country whose income is concentrated on firms whose costs are set by a single foreign country. The argument holds at any wage vector at which these groups are small, including the equilibrium one.

D Comparative Statics

D.1 Proof of Proposition 6. The first two equations are the market step of C.3, with the active sets and E fixed. If only j's costs change, $d\ln P = \varpi_j \hat{c}_j$, so $d\ln S_j = -(\sigma-1)\Lambda_j(1-\varpi_j)\hat{c}_j$ and, for $h \neq j$, $d\ln S_h = (\sigma-1)\Lambda_h \varpi_j \hat{c}_j$. Prices are $p_i = \mu(S_g)c_i$, so $d\ln p_i = \hat{c}_i + \varepsilon_\mu(S_g)\, d\ln S_g$, Averaging over group j's plants with weights s_i/S_j gives $\rho_j \hat{c}_j$, and every rival's price rises. Since $(\sigma-1)\varepsilon_\mu\Lambda_j < 1$ and $1 - \varpi_j \in [0,1)$, $\rho_j \in (0,1]$. Under quantity competition with $\eta = 1$, $\varepsilon_\mu = S/(1-S)$, which gives the stated Λ_g. □

D.2 Proof of Proposition 7. With $m(S) = (1+\kappa S)/\sigma$, $\pi^H/E = \sigma^{-1}\sum_g \theta_g S_g(1+\kappa S_g) = \sigma^{-1}[\bar{\theta} + \kappa \sum_g \theta_g S_g^2]$, and $\sum_g \theta_g S_g^2 = \bar{\theta}\cdot\mathrm{HHI} + \mathrm{Cov}_S(\theta, S)$ by definition. Total profit is the case $\theta_g \equiv 1$, $\sum_g \pi_g/E = (1+\kappa\,\mathrm{HHI})/\sigma$, using $\sum_g S_g = 1$. Subtracting $\bar{\theta}\sum_g \pi_g$ from π^H leaves $(E\kappa/\sigma)\mathrm{Cov}_S(\theta, S)$. The constant-markup benchmark is the case $\kappa = 0$. □

D.3 Proof of Proposition 8. By (14), $d\pi_g = (E/\sigma)(1+2\kappa S_g)\, dS_g$. Let $Z = \sum_h S_h \Lambda_h$, so $\varpi_j = S_j\Lambda_j/Z$. From D.1, $dS_j = -(\sigma-1)S_j\Lambda_j(1-\varpi_j)\hat{c}_j = -(\sigma-1)\varpi_j\hat{c}_j \sum_{h\neq j} S_h\Lambda_h$ and $dS_h = (\sigma-1)\varpi_j S_h \Lambda_h \hat{c}_j$ for $h \neq j$. Substituting into $d\pi^H = \sum_g \theta_g d\pi_g$ and collecting terms gives (22). □

D.4 Proof of Corollary 1. Monotonicity in A is Step 2 of B.3. For spending, $\Omega(K; A) = E\,\tilde{\Omega}(K; A)$ with $\tilde{\Omega}$ strictly increasing in K, so for two rungs with $K_1 < K_2$ the net gain $E[\tilde{\Omega}(K_2; A) - \tilde{\Omega}(K_1; A)] - [\Phi_2 - \Phi_1]$ is strictly increasing in E. So as E rises, no rung below the group's current choice can become optimal, and the chosen strength does not fall. □

E Solution Algorithm

E.1 Markets and entry. Given an aggregate A, each group's share solves the scalar equation (12), and the market solves (13) by safeguarded Newton steps on $\ln A$, which never leave a bracket that contains the unique root. With entry, each group's best response at a given A is found by scanning its ladder of efficient sets, and A is found by bisection.

E.2 Prices, incomes and wages. Given wages and active sets, price indices are obtained by iterating on (4)–(2), and market sizes and incomes by one linear solve. Wages are updated by the tâtonnement of Theorem 2, with a step within the bound in condition (ii), followed by Newton steps once close to the solution.

E.3 Active sets. At the equilibrium wages for the current configuration, the groups with the largest gains from changing their sets do so, and the procedure repeats until no group wishes to move. Moving only the groups with the largest gains changes the path of the algorithm, not its fixed points, and does not amount to an order of entry. When the loop does not reach a configuration in which no group wishes to move, it returns the configuration with the fewest such groups and reports how many there are. Because the joint equilibrium is not known to be unique, we solve from several starting wage vectors and compare the configurations reached.